



**MOTOROLA**  
**Semiconductors**

BOX 20912 • PHOENIX, ARIZONA 85036

**MC6800**

## Advance Information

### MICROPROCESSING UNIT (MPU)

The MC6800 is a monolithic 8-bit microprocessor forming the central control function for Motorola's M6800 family. Compatible with TTL, the MC6800, as with all M6800 system parts, requires only one +5.0-volt power supply, and no external TTL devices for bus interface.

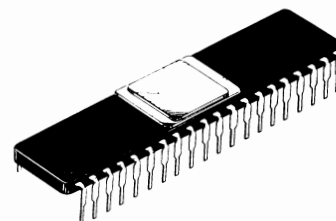
The MC6800 is capable of addressing 65 k bytes of memory with its 16-bit address lines. The 8-bit data bus is bidirectional as well as 3-state, making direct memory addressing and multiprocessing applications realizable.

- Eight-Bit Parallel Processing
- Bi-Directional Data Bus
- Sixteen-Bit Address Bus — 65 k Bytes of Addressing
- 72 Instructions — Variable Length
- Seven Addressing Modes — Direct, Relative, Immediate, Indexed, Extended, Implied and Accumulator
- Variable Length Stack
- Vectored Restart
- Maskable Interrupt Vector
- Separate Non-Maskable Interrupt — Internal Registers Saved In Stack
- Six Internal Registers — Two Accumulators, Index Register, Program Counter, Stack Pointer and Condition Code Register
- Direct Memory Addressing (DMA) and Multiple Processor Capability
- Clock Rates as High as 1 MHz
- Simple Bus Interface Without TTL
- Halt and Single Instruction Execution Capability

**MOS**

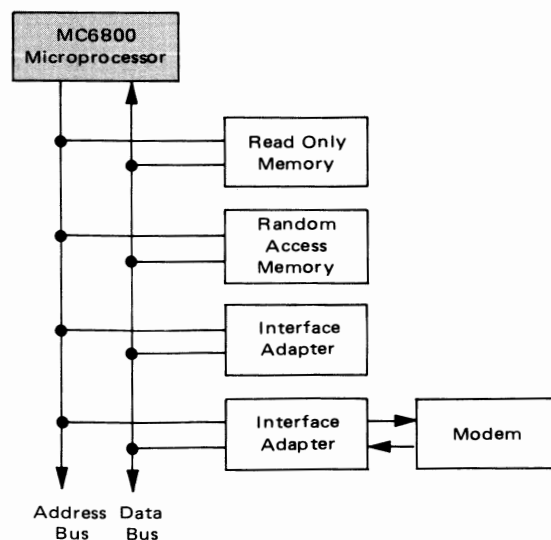
(N-CHANNEL, SILICON-GATE)

**MICROPROCESSOR**

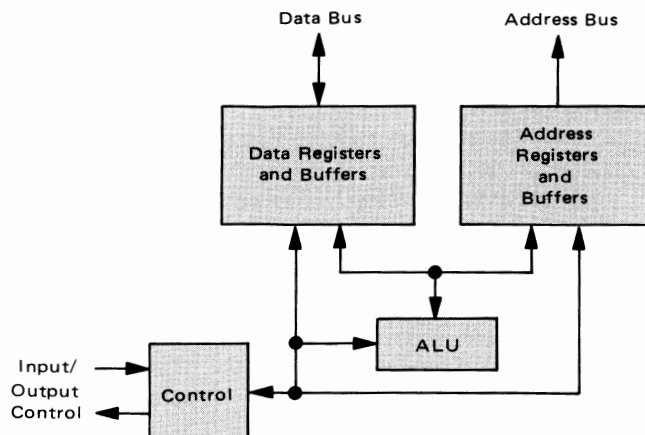


CERAMIC PACKAGE  
CASE 699

**M6800 MICROCOMPUTER FAMILY  
BLOCK DIAGRAM**



**MC6800 MICROPROCESSOR  
BLOCK DIAGRAM**



## MAXIMUM RATINGS

| Rating                      | Symbol    | Value        | Unit |
|-----------------------------|-----------|--------------|------|
| Supply Voltage              | $V_{CC}$  | -0.3 to +7.0 | Vdc  |
| Input Voltage               | $V_{in}$  | -0.3 to +7.0 | Vdc  |
| Operating Temperature Range | $T_A$     | 0 to +70     | °C   |
| Storage Temperature Range   | $T_{stg}$ | -55 to +150  | °C   |

This device contains circuitry to protect the inputs against damage due to high static voltages or electric fields; however, it is advised that normal precautions be taken to avoid application of any voltage higher than maximum rated voltages to this high impedance circuit.

ELECTRICAL CHARACTERISTICS ( $V_{CC} = 5.0 \text{ V} \pm 5\%$ ,  $V_{SS} = 0$ ,  $T_A = 25^\circ\text{C}$  unless otherwise noted.)

| Characteristic                                                                                         | Symbol                | Min                              | Typ           | Max                              | Unit            |
|--------------------------------------------------------------------------------------------------------|-----------------------|----------------------------------|---------------|----------------------------------|-----------------|
| Input High Voltage (Normal Operating Levels)                                                           |                       |                                  |               |                                  | Vdc             |
| Logic $\phi 1, \phi 2$                                                                                 | $V_{IH}$<br>$V_{IHC}$ | $V_{SS} + 2.4$<br>$V_{CC} - 0.3$ | —<br>—        | $V_{CC}$<br>$V_{CC} + 0.1$       |                 |
| Input Low Voltage (Normal Operating Levels)                                                            |                       |                                  |               |                                  | Vdc             |
| Logic $\phi 1, \phi 2$                                                                                 | $V_{IL}$<br>$V_{ILC}$ | $V_{SS} - 0.3$<br>$V_{SS} - 0.1$ | —<br>—        | $V_{SS} + 0.4$<br>$V_{SS} + 0.3$ |                 |
| Clock Overshoot/Undershoot — Input High Level<br>— Input Low Level                                     | $V_{OS}$              | $V_{CC} - 0.5$<br>$V_{SS} - 0.5$ | —<br>—        | $V_{CC} + 0.5$<br>$V_{SS} + 0.5$ | Vdc             |
| Input High Threshold Voltage<br>$\text{Reset}, \text{NMI}, \text{Halt}, \text{IRQ}, \text{Data}$       | $V_{IHT}$             | $V_{SS} + 2.0$                   | —             | —                                | Vdc             |
| Input Low Threshold Voltage<br>$\text{Reset}, \text{NMI}, \text{Halt}, \text{IRQ}, \text{Data}$        | $V_{ILT}$             | —                                | —             | $V_{SS} + 0.8$                   | Vdc             |
| Input Leakage Current<br>( $V_{in} = 0$ to $5.25 \text{ V}$ , $V_{CC} = 5.25 \text{ V}$ )              | $I_{in}$              | —<br>—                           | —<br>—        | 2.5<br>100                       | $\mu\text{Adc}$ |
| Three-State (Off State) Input Current<br>( $V_{in} = 0.4$ to $2.4 \text{ V}$ , $V_{CC} = \text{max}$ ) | $I_{TSI}$             | —<br>—                           | —<br>—        | 10<br>100                        | $\mu\text{Adc}$ |
| Data<br>A0-A15,R/W                                                                                     |                       |                                  |               |                                  |                 |
| Output High Voltage<br>( $I_{Load} = -100 \mu\text{Adc}$ , $V_{CC} = \text{min}$ )                     | $V_{OH}$              | $V_{SS} + 2.4$                   | —             | —                                | Vdc             |
| Output Low Voltage<br>( $I_{Load} = 1.6 \text{ mAdc}$ , $V_{CC} = \text{min}$ )                        | $V_{OL}$              | —                                | —             | $V_{SS} + 0.4$                   | Vdc             |
| Power Dissipation                                                                                      | $P_D$                 | —                                | 0.600         | 1.2                              | W               |
| Capacitance #<br>( $V_{in} = 0$ , $T_A = 25^\circ\text{C}$ , $f = 1.0 \text{ MHz}$ )                   | $C_{in}$              | —<br>—<br>80                     | —<br>—<br>120 | 10<br>15<br>160                  | pF              |
| Logic<br>Data,TSC<br>$\phi 1, \phi 2$<br>A0-A15,R/W                                                    |                       |                                  |               |                                  |                 |
| $C_{out}$                                                                                              |                       | —                                | —             | 12                               | pF              |
| Frequency of Operation                                                                                 | $f$                   | 0.1                              | —             | 1.0                              | MHz             |
| Clock Timing (Figure 1)                                                                                |                       |                                  |               |                                  |                 |
| Cycle Time                                                                                             | $t_{cyc}$             | 1.0                              | —             | 10                               | $\mu\text{s}$   |
| Clock Pulse Width<br>(Measured at $V_{CC} - 0.3 \text{ V}$ )                                           | $PW_{\phi H}$         |                                  |               |                                  | ns              |
| $\phi 1$                                                                                               |                       | 430                              | —             | 4500                             |                 |
| $\phi 2$                                                                                               |                       | 450                              | —             | 4500                             |                 |
| Total $\phi 1$ and $\phi 2$ Up Time                                                                    | $t_{ut}$              | 940                              | —             | —                                | ns              |
| Rise and Fall Times<br>(Measured between $V_{SS} + 0.3 \text{ V}$ and $V_{CC} - 0.3 \text{ V}$ )       | $t_r, t_f$            | 5.0                              | —             | 50                               | ns              |
| Delay Time or Clock Overlap<br>(Measured at $V_{SS} + 0.5 \text{ V}$ )                                 | $t_d$                 | 0                                | —             | 9100                             | ns              |
| Overshoot/Undershoot Duration                                                                          | $t_{OS}$              | 0                                | —             | 40                               | ns              |

\*Except  $\text{IRQ}$  and  $\text{NMI}$ , which require  $3 \text{ k}\Omega$  pullup load resistors for wire-OR capability at optimum operation.

# Capacitances are periodically sampled rather than 100% tested.



**READ/WRITE TIMING** Figures 2 and 3,  $f = 1.0 \text{ MHz}$ , Loading = 130 pF and one TTL Load except VMA and BA Loading = 30 pF and one TTL Load.

| Characteristic                                                   | Symbol    | Min | Typ | Max | Unit |
|------------------------------------------------------------------|-----------|-----|-----|-----|------|
| Read/Write Setup Time from MPU                                   | $T_{ASR}$ | —   | 100 | 300 | ns   |
| Address Setup Time from MPU                                      | $T_{ASC}$ | —   | 200 | 300 | ns   |
| Memory Read Access Time<br>$t_{cyc} - (T_{ASC} + T_{DSU} + t_r)$ | $T_{ACC}$ | —   | —   | 575 | ns   |
| Data Setup Time                                                  | $T_{DSU}$ | 100 | —   | —   | ns   |
| Address Setup Time from MPU for VMA                              | $T_{VSC}$ | —   | 150 | 300 | ns   |
| Data Hold Time                                                   | $T_H$     | 10  | 30  | —   | ns   |
| Enable High Time for DBE Input                                   | $T_{EH}$  | 470 | —   | —   | μs   |
| Data Setup Time from MPU                                         | $T_{ASD}$ | —   | 150 | 200 | ns   |

FIGURE 1 – CLOCK TIMING WAVEFORM

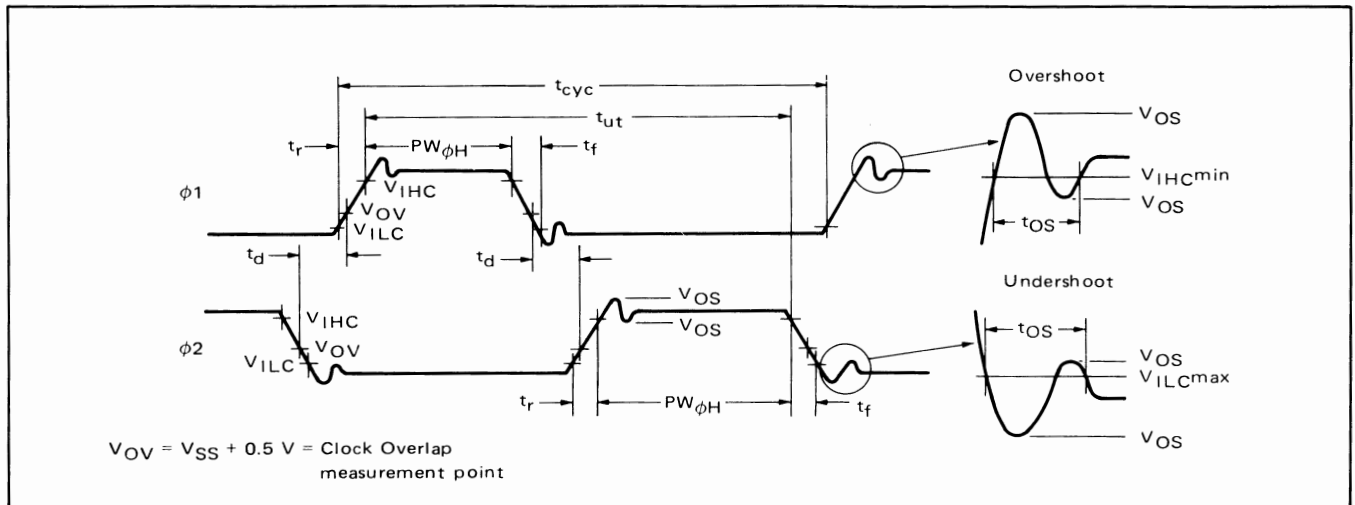


FIGURE 2 – READ DATA FROM MEMORY OR PERIPHERALS

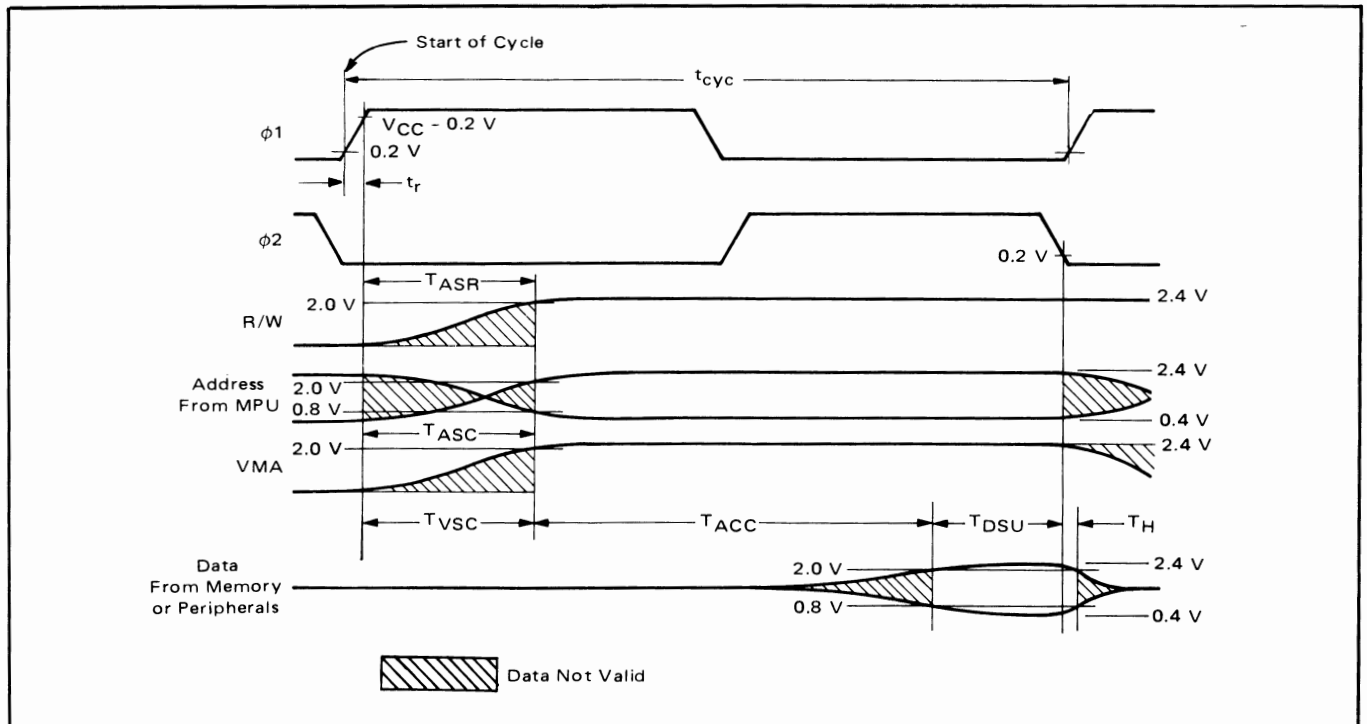
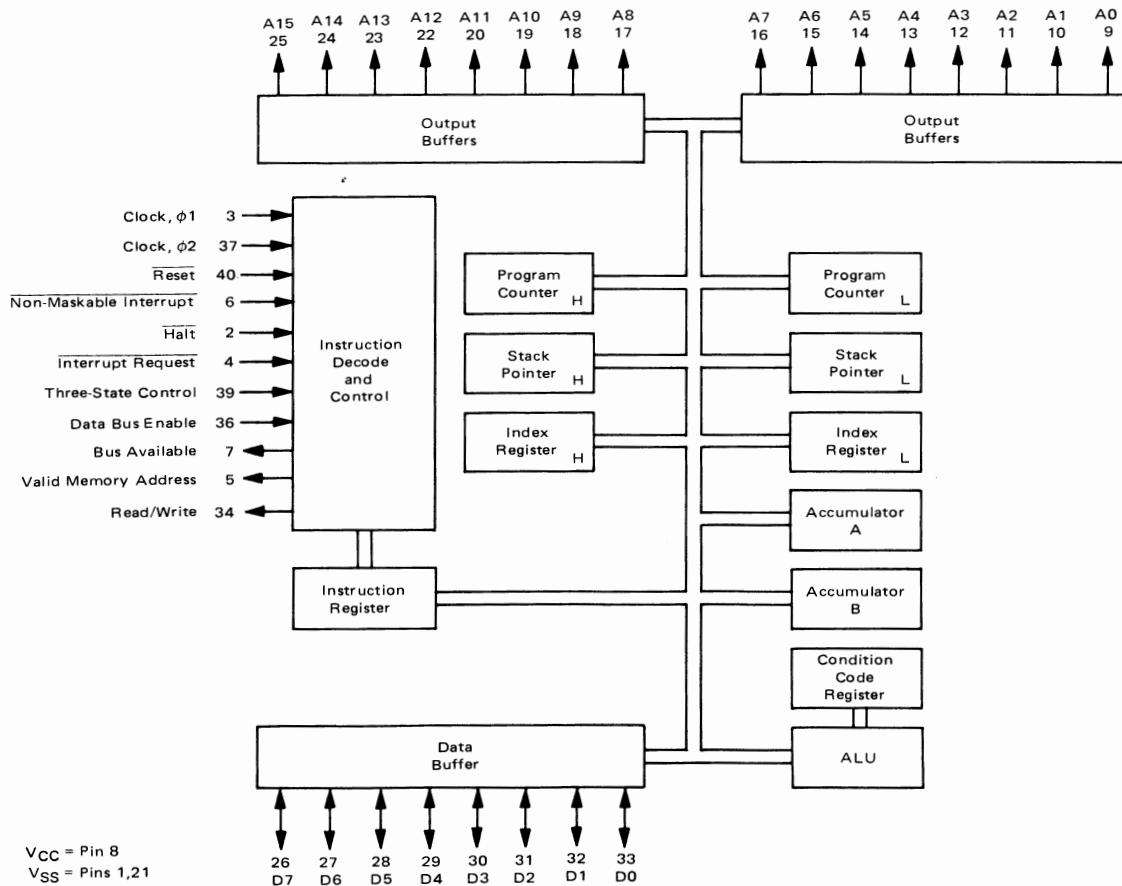
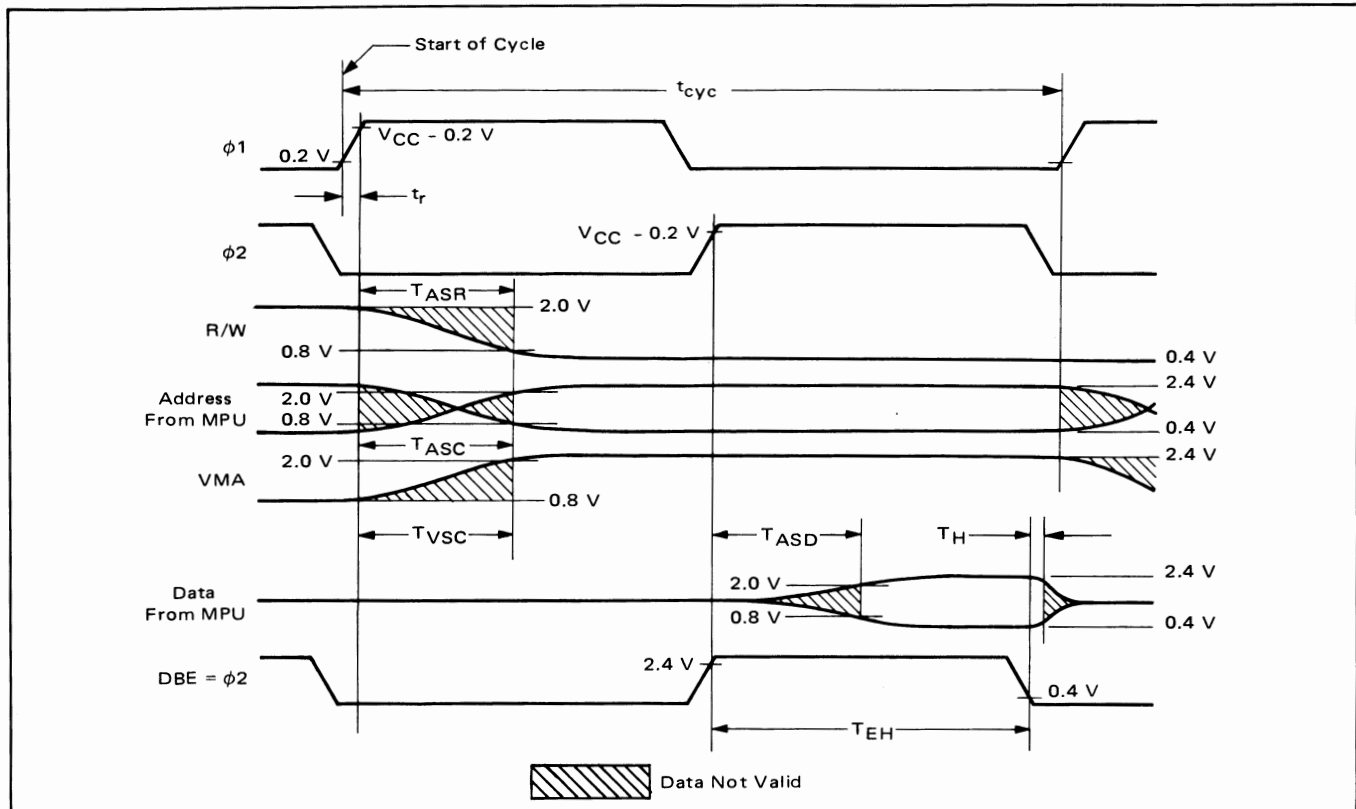


FIGURE 3 – WRITE DATA IN MEMORY OR PERIPHERALS



## MPU SIGNAL DESCRIPTION

Proper operation of the MPU requires that certain control and timing signals be provided to accomplish specific functions and that other signal lines be monitored to determine the state of the processor.

**Clocks Phase One and Phase Two ( $\phi 1, \phi 2$ )** — Two pins are used for a two-phase non-overlapping clock that runs at the  $V_{CC}$  voltage level.

**Address Bus (A0-A15)** — Sixteen pins are used for the address bus. The outputs are three-state bus drivers capable of driving one standard TTL load and 130 pF. When the output is turned off, it is essentially an open circuit. This permits the MPU to be used in DMA applications.

**Data Bus (D0-D7)** — Eight pins are used for the data bus. It is bi-directional, transferring data to and from the memory and peripheral devices. It also has three-state output buffers capable of driving one standard TTL load and 130 pF.

**Halt** — When this input is in the low state, all activity in the machine will be halted. This input is level sensitive. In the halt mode, the machine will stop at the end of an instruction, Bus Available will be at a one level, Valid Memory Address will be at a zero, and all other three-state lines will be in the three-state mode.

Transition of the  $\overline{\text{Halt}}$  line must not occur during the last 250 ns of phase one. To insure single instruction operation, the  $\overline{\text{Halt}}$  line must go high for one Phase One Clock cycle.

**Three-State Control (TSC)** — This input causes all of the address lines and the Read/Write line to go into the off or high impedance state. This state will occur 500 ns after  $TSC = 2.4 V$ . The Valid Memory Address and Bus Available signals will be forced low. The data bus is not affected by TSC and has its own enable (Data Bus Enable). In DMA applications, the Three-State Control line should be brought high on the leading edge of the Phase One Clock. The  $\phi 1$  clock must be held in the high state and the  $\phi 2$  in the low state for this function to operate properly. The address bus will then be available for other devices to directly address memory. Since the MPU is a dynamic device, it can be held in this state for only 5.0  $\mu s$  or destruction of data will occur in the MPU.

**Read/Write (R/W)** — This TTL compatible output signals the peripherals and memory devices whether the MPU is in a Read (high) or Write (low) state. The normal standby state of this signal is Read (high). Three-State Control going high will turn Read/Write to the off (high impedance) state. Also, when the processor is halted, it will be in the off state. This output is capable of driving one standard TTL load and 130 pF.

**Valid Memory Address (VMA)** — This output indicates to peripheral devices that there is a valid address on the address bus. In normal operation, this signal should be utilized for enabling peripheral interfaces such as the PIA and ACIA. This signal is not three-state. One standard TTL load and 30 pF may be directly driven by this active high signal.

**Data Bus Enable (DBE)** — This input is the three-state control signal for the MPU data bus and will enable the bus drivers when in the high state. This input is TTL compatible; however in normal operation, it would be driven by the phase two clock. During an MPU read cycle, the data bus drivers will be disabled internally. When it is desired that another device control the data bus such as in Direct Memory Access (DMA) applications, DBE should be held low.

**Bus Available (BA)** — The Bus Available signal will normally be in the low state; when activated, it will go to the high state indicating that the microprocessor has stopped and that the address bus is available. This will occur if the  $\overline{\text{Halt}}$  line is in the low state or the processor is in the WAIT state as a result of the execution of a WAIT instruction. At such time, all three-state output drivers will go to their off state and other outputs to their normally inactive level. The processor is removed from the WAIT state by the occurrence of a maskable (mask bit  $I = 0$ ) or nonmaskable interrupt. This output is capable of driving one standard TTL load and 30 pF.

**Interrupt Request ( $\overline{\text{IRQ}}$ )** — This level sensitive input requests that an interrupt sequence be generated within the machine. The processor will wait until it completes the current instruction that is being executed before it recognizes the request. At that time, if the interrupt mask bit in the Condition Code Register is not set, the machine will begin an interrupt sequence. The Index Register, Program Counter, Accumulators, and Condition Code Register are stored away on the stack. Next the MPU will respond to the interrupt request by setting the interrupt mask bit high so that no further interrupts may occur. At the end of the cycle, a 16-bit address will be loaded that points to a vectoring address which is located in memory locations FFF8 and FFF9. An address loaded at these locations causes the MPU to branch to an interrupt routine in memory.

The  $\overline{\text{Halt}}$  line must be in the high state for interrupts to be recognized.

The  $\overline{\text{IRQ}}$  has a high impedance pullup device internal to the chip; however a 3 k $\Omega$  external resistor to  $V_{CC}$  should be used for wire-OR and optimum control of interrupts.

**Reset** — This input is used to reset and start the MPU from a power down condition, resulting from a power failure or an initial start-up of the processor. If a positive edge is detected on the input, this will signal the MPU to begin the restart sequence. This will start execution of a routine to initialize the processor from its reset condition. All the higher order address lines will be forced high. For the restart, the last two (FFFE, FFFF) locations in memory will be used to load the program that is addressed by the program counter. During the restart routine, the interrupt mask bit is set and must be reset before the MPU can be interrupted by  $\overline{\text{IRQ}}$ .



Figure 4 shows the initialization of the microprocessor after restart.  $\overline{\text{Reset}}$  must be held low for at least eight clock periods after  $V_{CC}$  reaches 4.75 volts. If  $\overline{\text{Reset}}$  goes high prior to the leading edge of  $\phi_2$ , on the next  $\phi_1$  the first restart memory vector address (FFFE) will appear on the address lines. This location should contain the higher order eight bits to be stored into the program counter. Following, the next address FFFF should contain the lower order eight bits to be stored into the program counter.

**Non-Maskable Interrupt (NMI)** — A low-going edge on this input requests that a non-mask-interrupt sequence be generated within the processor. As with the Interrupt Request signal, the processor will complete the current instruction that is being executed before it recognizes the  $\overline{\text{NMI}}$  signal. The interrupt mask bit in the Condition Code Register has no effect on  $\overline{\text{NMI}}$ .

The Index Register, Program Counter, Accumulators, and Condition Code Register are stored away on the stack. At the end of the cycle, a 16-bit address will be loaded that points to a vectored address which is located in memory locations FFFC and FFFD. An address loaded at these locations causes the MPU to branch to a non-maskable interrupt routine in memory.

NMI has a high impedance pullup resistor internal to the chip; however a 3 k $\Omega$  external resistor to  $V_{CC}$  should be used for wire-OR and optimum control of interrupts.

Inputs  $\overline{\text{IRQ}}$  and  $\overline{\text{NMI}}$  are hardware interrupt lines that are sampled during  $\phi_2$  and will start the interrupt routine on the  $\phi_1$  following the completion of an instruction.

Figure 5 is a flow chart describing the major decision paths and interrupt vectors of the microprocessor. Table 1 gives the memory map for interrupt vectors.

FIGURE 4 — INITIALIZATION OF MPU AFTER RESTART

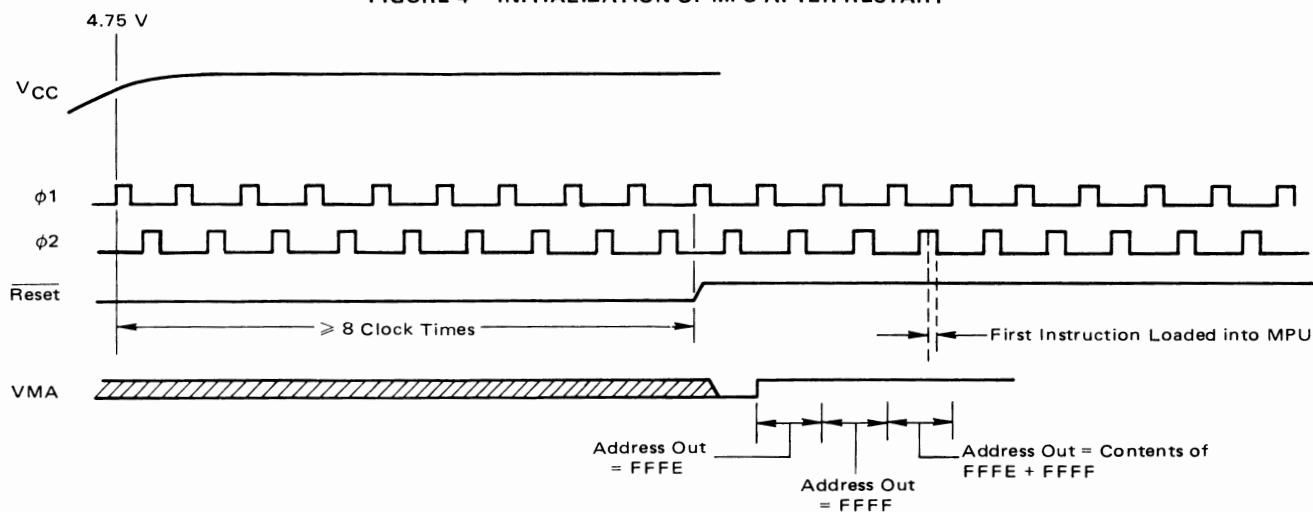
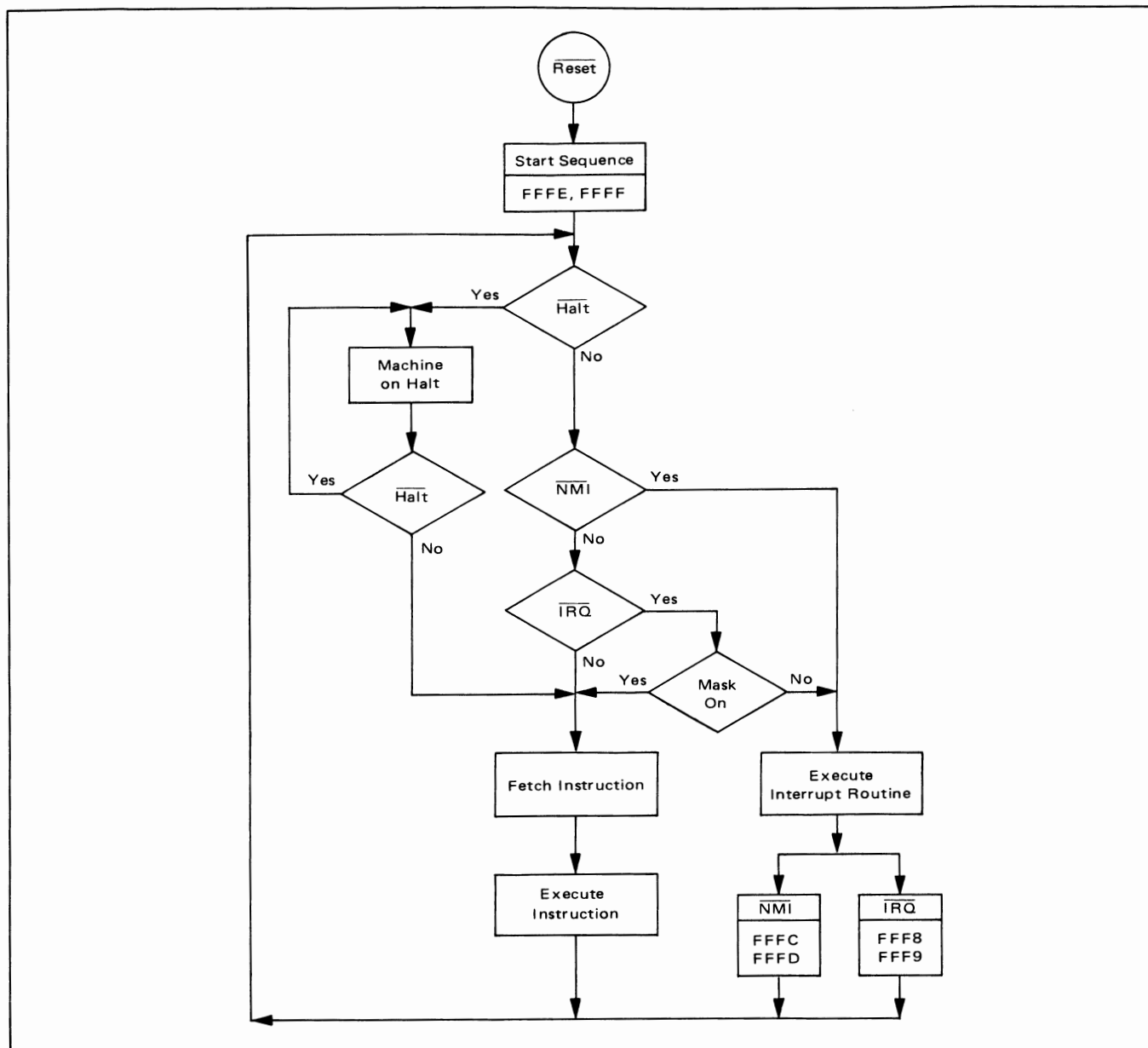


TABLE 1 — MEMORY MAP FOR INTERRUPT VECTORS

| Vector |      | Description            |
|--------|------|------------------------|
| MS     | LS   |                        |
| FFFE   | FFFF | Restart                |
| FFFC   | FFFD | Non-maskable Interrupt |
| FFFA   | FFFB | Software Interrupt     |
| FFF8   | FFF9 | Interrupt Request      |



FIGURE 5 – MPU FLOW CHART



## MPU REGISTERS

The MPU has three 16-bit registers and three 8-bit registers available for use by the programmer (Figure 6).

**Program Counter** — The program counter is a two byte (16-bits) register that points to the current program address.

**Stack Pointer** — The stack pointer is a two byte register that contains the address of the next available location in an external push-down/pop-up stack. This stack is normally a random access Read/Write memory that may

have any location (address) that is convenient. In those applications that require storage of information in the stack when power is lost, the stack must be non-volatile.

**Index Register** — The index register is a two byte register that is used to store data or a sixteen bit memory address for the Indexed mode of memory addressing.

**Accumulators** — The MPU contains two 8-bit accumulators that are used to hold operands and results from an arithmetic logic unit (ALU).



FIGURE 6 – PROGRAMMING MODEL OF THE MICROPROCESSING UNIT

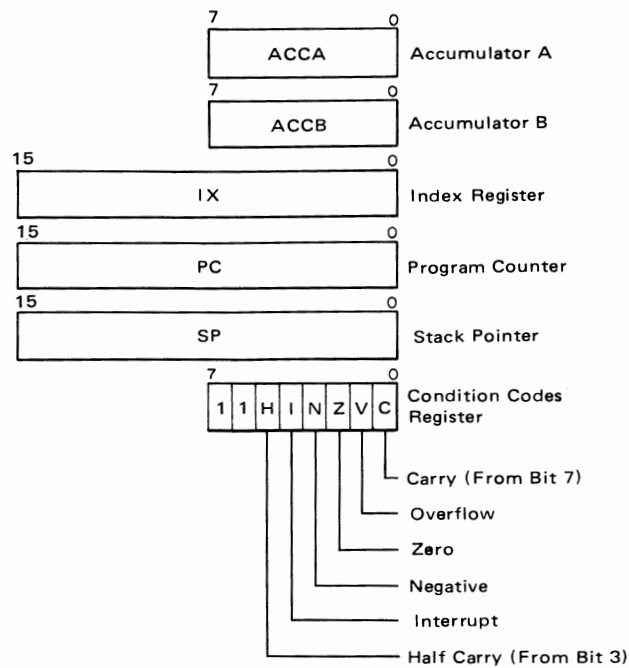
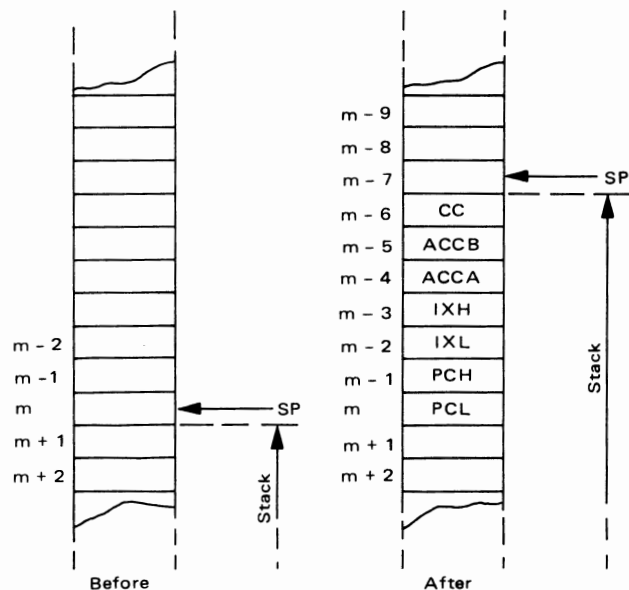


FIGURE 7 – SAVING THE STATUS OF THE MICROPROCESSOR IN THE STACK

SP = Stack Pointer  
 CC = Condition Codes (Also called the Processor Status Byte)  
 ACCB = Accumulator B  
 ACCA = Accumulator A  
 IXH = Index Register, Higher Order 8 Bits  
 IXL = Index Register, Lower Order 8 Bits  
 PCH = Program Counter, Higher Order 8 Bits  
 PCL = Program Counter, Lower Order 8 Bits





**Condition Code Register** — The condition code register indicates the results of an Arithmetic Logic Unit operation: Negative (N), Zero (Z), Overflow (V), Carry from bit 7 (C), and half carry from bit 3 (H). These bits of the Condition Code Register are used as testable conditions for the conditional branch instructions. Bit 4 is the interrupt mask bit (I). The unused bits of the Condition Code Register (b6 and b7) are ones.

Figure 8 shows the order of saving the microprocessor status within the stack.

## MPU INSTRUCTION SET

The MC6800 has a set of 72 different instructions. Included are binary and decimal arithmetic, logical, shift, rotate, load, store, conditional or unconditional branch, interrupt and stack manipulation instructions (Tables 2 thru 6).

## MPU ADDRESSING MODES

The MC6800 eight-bit microprocessing unit has seven address modes that can be used by a programmer, with the addressing mode a function of both the type of instruction and the coding within the instruction. A summary of the addressing modes for a particular instruction can be found in Table 7 along with the associated instruction execution time that is given in machine cycles. With a clock frequency of 1 MHz, these times would be microseconds.

**Accumulator (ACCX) Addressing** — In accumulator only addressing, either accumulator A or accumulator B is specified. These are one-byte instructions.

**Immediate Addressing** — In immediate addressing, the operand is contained in the second byte of the instruction except LDS and LDX which have the operand in the second and third bytes of the instruction. The MPU addresses

this location when it fetches the immediate instruction for execution. These are two or three-byte instructions.

**Direct Addressing** — In direct addressing, the address of the operand is contained in the second byte of the instruction. Direct addressing allows the user to directly address the lowest 256 bytes in the machine i.e., locations zero through 255. Enhanced execution times are achieved by storing data in these locations. In most configurations, it should be a random access memory. These are two-byte instructions.

**Extended Addressing** — In extended addressing, the address contained in the second byte of the instruction is used as the higher eight-bits of the address of the operand. The third byte of the instruction is used as the lower eight-bits of the address for the operand. This is an absolute address in memory. These are three-byte instructions.

**Indexed Addressing** — In indexed addressing, the address contained in the second byte of the instruction is added to the index register's lowest eight bits in the MPU. The carry is then added to the higher order eight bits of the index register. This result is then used to address memory. The modified address is held in a temporary address register so there is no change to the index register. These are two-byte instructions.

**Implied Addressing** — In the implied addressing mode the instruction gives the address (i.e., stack pointer, index register, etc.). These are one-byte instructions.

**Relative Addressing** — In relative addressing, the address contained in the second byte of the instruction is added to the program counter's lowest eight bits plus two. The carry or borrow is then added to the high eight bits. This allows the user to address data within a range of -125 to +129 bytes of the present instruction. These are two-byte instructions.

TABLE 2 — MICROPROCESSOR INSTRUCTION SET — ALPHABETIC SEQUENCE

|     |                                 |     |                          |     |                                              |
|-----|---------------------------------|-----|--------------------------|-----|----------------------------------------------|
| ABA | Add Accumulators                | CLR | Clear                    | PUL | Pull Data                                    |
| ADC | Add with Carry                  | CLV | Clear Overflow           | ROL | Rotate Left                                  |
| ADD | Add                             | CMP | Compare                  | ROR | Rotate Right                                 |
| AND | Logical And                     | COM | Complement               | RTI | Return from Interrupt                        |
| ASL | Arithmetic Shift Left           | CPX | Compare Index Register   | RTS | Return from Subroutine                       |
| ASR | Arithmetic Shift Right          | DAA | Decimal Adjust           | SBA | Subtract Accumulators                        |
| BCC | Branch if Carry Clear           | DEC | Decrement                | SBC | Subtract with Carry                          |
| BCS | Branch if Carry Set             | DES | Decrement Stack Pointer  | SEC | Set Carry                                    |
| BEQ | Branch if Equal to Zero         | DEX | Decrement Index Register | SEI | Set Interrupt Mask                           |
| BGE | Branch if Greater or Equal Zero | EOR | Exclusive OR             | SEV | Set Overflow                                 |
| BGT | Branch if Greater than Zero     | INC | Increment                | STA | Store Accumulator                            |
| BHI | Branch if Higher                | INS | Increment Stack Pointer  | STS | Store Stack Register                         |
| BIT | Bit Test                        | INX | Increment Index Register | STX | Store Index Register                         |
| BLE | Branch if Less or Equal         | JMP | Jump                     | SUB | Subtract                                     |
| BLS | Branch if Lower or Same         | JSR | Jump to Subroutine       | SWI | Software Interrupt                           |
| BLT | Branch if Less than Zero        | LDA | Load Accumulator         | TAB | Transfer Accumulators                        |
| BMI | Branch if Minus                 | LDS | Load Stack Pointer       | TAP | Transfer Accumulators to Condition Code Reg. |
| BNE | Branch if Not Equal to Zero     | LDX | Load Index Register      | TBA | Transfer Accumulators                        |
| BPL | Branch if Plus                  | LSR | Logical Shift Right      | TPA | Transfer Condition Code Reg. to Accumulator  |
| BRA | Branch Always                   | NEG | Negate                   | TST | Test                                         |
| BSR | Branch to Subroutine            | NOP | No Operation             | TSX | Transfer Stack Pointer to Index Register     |
| BVC | Branch if Overflow Clear        | ORA | Inclusive OR Accumulator | TXS | Transfer Index Register to Stack Pointer     |
| BVS | Branch if Overflow Set          | PSH | Push Data                | WAI | Wait for Interrupt                           |
| CBA | Compare Accumulators            |     |                          |     |                                              |
| CLC | Clear Carry                     |     |                          |     |                                              |
| CLI | Clear Interrupt Mask            |     |                          |     |                                              |



TABLE 3 – ACCUMULATOR AND MEMORY INSTRUCTIONS

| ADDRESSING MODES         |          |       |     |        |     |       |     |       |     | BOOLEAN/ARITHMETIC OPERATION |     |                                                        |   | COND. CODE REG. |   |   |   |    |
|--------------------------|----------|-------|-----|--------|-----|-------|-----|-------|-----|------------------------------|-----|--------------------------------------------------------|---|-----------------|---|---|---|----|
| OPERATIONS               | MNEMONIC | IMMED |     | DIRECT |     | INDEX |     | EXTND |     | IMPLIED                      |     | (All register labels refer to contents)                | 5 | 4               | 3 | 2 | 1 | 0  |
|                          |          | OP    | ~ = | OP     | ~ = | OP    | ~ = | OP    | ~ = | OP                           | ~ = |                                                        | H | I               | N | Z | V | C  |
| Add                      | ADDA     | 8B    | 2 2 | 9B     | 3 2 | AB    | 5 2 | BB    | 4 3 |                              |     | A ← M → A                                              | ↑ | ●               | ↑ | ↑ | ↑ | ↑  |
|                          | ADDB     | CB    | 2 2 | DB     | 3 2 | EB    | 5 2 | FB    | 4 3 |                              |     | B ← M → B                                              | ↑ | ●               | ↑ | ↑ | ↑ | ↑  |
| Add Acmltrs              | ABA      |       |     |        |     |       |     |       |     | 1B                           | 2 1 | A ← B → A                                              | ↑ | ●               | ↑ | ↑ | ↑ | ↑  |
| Add with Carry           | ADCA     | 89    | 2 2 | 99     | 3 2 | A9    | 5 2 | B9    | 4 3 |                              |     | A ← M + C → A                                          | ↑ | ↑               | ↑ | ↑ | ↑ | ↑  |
|                          | ADCB     | C9    | 2 2 | D9     | 3 2 | E9    | 5 2 | F9    | 4 3 |                              |     | B ← M + C → B                                          | ↑ | ↑               | ↑ | ↑ | ↑ | ↑  |
| And                      | ANDA     | 84    | 2 2 | 94     | 3 2 | A4    | 5 2 | B4    | 4 3 |                              |     | A ← M → A                                              | ● | ↑               | ↑ | ↑ | ↑ | R  |
|                          | ANDB     | C4    | 2 2 | D4     | 3 2 | E4    | 5 2 | F4    | 4 3 |                              |     | B ← M → B                                              | ● | ↑               | ↑ | ↑ | ↑ | R  |
| Bit Test                 | BITA     | 85    | 2 2 | 95     | 3 2 | A5    | 5 2 | B5    | 4 3 |                              |     | A ← M                                                  | ● | ↑               | ↑ | ↑ | ↑ | R  |
|                          | BITB     | C5    | 2 2 | D5     | 3 2 | E5    | 5 2 | F5    | 4 3 |                              |     | B ← M                                                  | ● | ↑               | ↑ | ↑ | ↑ | R  |
| Clear                    | CLR      |       |     |        |     | 6F    | 7 2 | 7F    | 6 3 |                              |     | 00 → M                                                 | ● | ●               | R | S | R | R  |
|                          | CLRA     |       |     |        |     |       |     |       |     | 4F                           | 2 1 | 00 → A                                                 | ● | ●               | R | S | R | R  |
|                          | CLRB     |       |     |        |     |       |     |       |     | 5F                           | 2 1 | 00 → B                                                 | ● | ●               | R | S | R | R  |
| Compare                  | CMPA     | 81    | 2 2 | 91     | 3 2 | A1    | 5 2 | B1    | 4 3 |                              |     | A ← M                                                  | ● | ↑               | ↑ | ↑ | ↑ | ↑  |
|                          | CMPB     | C1    | 2 2 | D1     | 3 2 | E1    | 5 2 | F1    | 4 3 |                              |     | B ← M                                                  | ● | ↑               | ↑ | ↑ | ↑ | ↑  |
| Compare Acmltrs          | CBA      |       |     |        |     |       |     |       |     | 11                           | 2 1 | A ← B                                                  | ● | ↑               | ↑ | ↑ | ↑ | ↑  |
| Complement, 1's          | COM      |       |     |        |     | 63    | 7 2 | 73    | 6 3 |                              |     | M → M                                                  | ● | ↑               | ↑ | ↑ | ↑ | R  |
|                          | COMA     |       |     |        |     |       |     |       |     | 43                           | 2 1 | A ← A                                                  | ● | ↑               | ↑ | ↑ | ↑ | R  |
|                          | COMB     |       |     |        |     |       |     |       |     | 53                           | 2 1 | B ← B                                                  | ● | ↑               | ↑ | ↑ | ↑ | R  |
| Complement, 2's (Negate) | NEG      |       |     |        |     | 60    | 7 2 | 70    | 6 3 |                              |     | 00 ← M → M                                             | ● | ↑               | ↑ | ↑ | ↑ | ①② |
|                          | NEGA     |       |     |        |     |       |     |       |     | 40                           | 2 1 | 00 ← A → A                                             | ● | ↑               | ↑ | ↑ | ↑ | ①② |
|                          | NEGB     |       |     |        |     |       |     |       |     | 50                           | 2 1 | 00 ← B → B                                             | ● | ↑               | ↑ | ↑ | ↑ | ①② |
| Decimal Adjust, A        | DAA      |       |     |        |     |       |     |       |     | 19                           | 2 1 | Converts Binary Add. of BCD Characters into BCD Format | ● | ↑               | ↑ | ↑ | ↑ | ③  |
| Decrement                | DEC      |       |     |        |     | 6A    | 7 2 | 7A    | 6 3 |                              |     | M ← 1 → M                                              | ● | ↑               | ↑ | ↑ | 4 | ●  |
|                          | DECA     |       |     |        |     |       |     |       |     | 4A                           | 2 1 | A ← 1 → A                                              | ● | ↑               | ↑ | ↑ | 4 | ●  |
|                          | DECB     |       |     |        |     |       |     |       |     | 5A                           | 2 1 | B ← 1 → B                                              | ● | ↑               | ↑ | ↑ | 4 | ●  |
| Exclusive OR             | EORA     | 88    | 2 2 | 98     | 3 2 | A8    | 5 2 | B8    | 4 3 |                              |     | A ⊕ M → A                                              | ● | ↑               | ↑ | ↑ | ↑ | R  |
|                          | EORB     | C8    | 2 2 | D8     | 3 2 | E8    | 5 2 | F8    | 4 3 |                              |     | B ⊕ M → B                                              | ● | ↑               | ↑ | ↑ | ↑ | R  |
| Increment                | INC      |       |     |        |     | 6C    | 7 2 | 7C    | 6 3 |                              |     | M ← 1 → M                                              | ● | ↑               | ↑ | ↑ | ↑ | ⑤  |
|                          | INCA     |       |     |        |     |       |     |       |     | 4C                           | 2 1 | A ← 1 → A                                              | ● | ↑               | ↑ | ↑ | ↑ | ⑤  |
|                          | INCB     |       |     |        |     |       |     |       |     | 5C                           | 2 1 | B ← 1 → B                                              | ● | ↑               | ↑ | ↑ | ↑ | ⑤  |
| Load Acmltr              | LDAA     | 86    | 2 2 | 96     | 3 2 | A6    | 5 2 | B6    | 4 3 |                              |     | M → A                                                  | ● | ↑               | ↑ | ↑ | ↑ | R  |
|                          | LDAB     | C6    | 2 2 | D6     | 3 2 | E6    | 5 2 | F6    | 4 3 |                              |     | M → B                                                  | ● | ↑               | ↑ | ↑ | ↑ | R  |
| Or, Inclusive            | ORAA     | 8A    | 2 2 | 9A     | 3 2 | AA    | 5 2 | BA    | 4 3 |                              |     | A ← M → A                                              | ● | ↑               | ↑ | ↑ | ↑ | R  |
|                          | ORAB     | CA    | 2 2 | DA     | 3 2 | EA    | 5 2 | FA    | 4 3 |                              |     | B ← M → B                                              | ● | ↑               | ↑ | ↑ | ↑ | R  |
| Push Data                | PSHA     |       |     |        |     |       |     |       |     | 36                           | 4 1 | A → M <sub>SP</sub> , SP ← 1 → SP                      | ● | ↑               | ↑ | ↑ | ↑ | ●  |
|                          | PSHB     |       |     |        |     |       |     |       |     | 37                           | 4 1 | B → M <sub>SP</sub> , SP ← 1 → SP                      | ● | ↑               | ↑ | ↑ | ↑ | ●  |
| Pull Data                | PULA     |       |     |        |     |       |     |       |     | 32                           | 4 1 | SP ← 1 → SP, M <sub>SP</sub> → A                       | ● | ↑               | ↑ | ↑ | ↑ | ●  |
|                          | PULB     |       |     |        |     |       |     |       |     | 33                           | 4 1 | SP ← 1 → SP, M <sub>SP</sub> → B                       | ● | ↑               | ↑ | ↑ | ↑ | ●  |
| Rotate Left              | ROL      |       |     |        |     | 69    | 7 2 | 79    | 6 3 |                              |     | M                                                      | ● | ↑               | ↑ | ↑ | ↑ | ⑥  |
|                          | ROLA     |       |     |        |     |       |     |       |     | 49                           | 2 1 | A                                                      | ● | ↑               | ↑ | ↑ | ↑ | ⑥  |
|                          | ROLB     |       |     |        |     |       |     |       |     | 59                           | 2 1 | B                                                      | ● | ↑               | ↑ | ↑ | ↑ | ⑥  |
| Rotate Right             | ROR      |       |     |        |     | 66    | 7 2 | 76    | 6 3 |                              |     | M                                                      | ● | ↑               | ↑ | ↑ | ↑ | ⑥  |
|                          | RORA     |       |     |        |     |       |     |       |     | 46                           | 2 1 | A                                                      | ● | ↑               | ↑ | ↑ | ↑ | ⑥  |
|                          | RORB     |       |     |        |     |       |     |       |     | 56                           | 2 1 | B                                                      | ● | ↑               | ↑ | ↑ | ↑ | ⑥  |
| Shift Left, Arithmetic   | ASL      |       |     |        |     | 68    | 7 2 | 78    | 6 3 |                              |     | M                                                      | ● | ↑               | ↑ | ↑ | ↑ | ⑥  |
|                          | ASLA     |       |     |        |     |       |     |       |     | 48                           | 2 1 | A                                                      | ● | ↑               | ↑ | ↑ | ↑ | ⑥  |
|                          | ASLB     |       |     |        |     |       |     |       |     | 58                           | 2 1 | B                                                      | ● | ↑               | ↑ | ↑ | ↑ | ⑥  |
| Shift Right, Arithmetic  | ASR      |       |     |        |     | 67    | 7 2 | 77    | 6 3 |                              |     | M                                                      | ● | ↑               | ↑ | ↑ | ↑ | ⑥  |
|                          | ASRA     |       |     |        |     |       |     |       |     | 47                           | 2 1 | A                                                      | ● | ↑               | ↑ | ↑ | ↑ | ⑥  |
|                          | ASRB     |       |     |        |     |       |     |       |     | 57                           | 2 1 | B                                                      | ● | ↑               | ↑ | ↑ | ↑ | ⑥  |
| Shift Right, Logic       | LSR      |       |     |        |     | 64    | 7 2 | 74    | 6 3 |                              |     | M                                                      | ● | ↑               | ↑ | ↑ | ↑ | ⑥  |
|                          | LSRA     |       |     |        |     |       |     |       |     | 44                           | 2 1 | A                                                      | ● | ↑               | ↑ | ↑ | ↑ | ⑥  |
|                          | LSRB     |       |     |        |     |       |     |       |     | 54                           | 2 1 | B                                                      | ● | ↑               | ↑ | ↑ | ↑ | ⑥  |
| Store Acmltr.            | STAA     |       |     | 97     | 4 2 | A7    | 6 2 | B7    | 5 3 |                              |     | A → M                                                  | ● | ↑               | ↑ | ↑ | ↑ | R  |
|                          | STAB     |       |     | 07     | 4 2 | E7    | 6 2 | F7    | 5 3 |                              |     | B → M                                                  | ● | ↑               | ↑ | ↑ | ↑ | R  |
| Subtract                 | SUBA     | 80    | 2 2 | 90     | 3 2 | A0    | 5 2 | B0    | 4 3 |                              |     | A ← M → A                                              | ● | ↑               | ↑ | ↑ | ↑ | ↑  |
|                          | SUBB     | C0    | 2 2 | D0     | 3 2 | E0    | 5 2 | F0    | 4 3 |                              |     | B ← M → B                                              | ● | ↑               | ↑ | ↑ | ↑ | ↑  |
| Subtract Acmltrs.        | SBA      |       |     |        |     |       |     |       |     | 10                           | 2 1 | A ← B → A                                              | ● | ↑               | ↑ | ↑ | ↑ | ↑  |
| Subtr. with Carry        | SBCA     | 82    | 2 2 | 92     | 3 2 | A2    | 5 2 | B2    | 4 3 |                              |     | A ← M - C → A                                          | ● | ↑               | ↑ | ↑ | ↑ | ↑  |
|                          | SBCB     | C2    | 2 2 | D2     | 3 2 | E2    | 5 2 | F2    | 4 3 |                              |     | B ← M - C → B                                          | ● | ↑               | ↑ | ↑ | ↑ | ↑  |
| Transfer Acmltrs         | TAB      |       |     |        |     |       |     |       |     | 16                           | 2 1 | A → B                                                  | ● | ↑               | ↑ | ↑ | ↑ | R  |
|                          | TBA      |       |     |        |     |       |     |       |     | 17                           | 2 1 | B → A                                                  | ● | ↑               | ↑ | ↑ | ↑ | R  |
| Test, Zero or Minus      | TST      |       |     |        |     | 6D    | 7 2 | 7D    | 6 3 |                              |     | M ← 00                                                 | ● | ↑               | ↑ | ↑ | ↑ | R  |
|                          | TSTA     |       |     |        |     |       |     |       |     | 4D                           | 2 1 | A ← 00                                                 | ● | ↑               | ↑ | ↑ | ↑ | R  |
|                          | TSTB     |       |     |        |     |       |     |       |     | 5D                           | 2 1 | B ← 00                                                 | ● | ↑               | ↑ | ↑ | ↑ | R  |

## LEGEND:

OP Operation Code (Hexadecimal);  
 ~ Number of MPU Cycles;  
 = Number of Program Bytes;  
 + Arithmetic Plus;  
 - Arithmetic Minus;  
 \* Boolean AND;

MSP Contents of memory location pointed to be Stack Pointer;

Note – Accumulator addressing mode instructions are included in the column for IMPLIED addressing

+ Boolean Inclusive OR;  
 ⊕ Boolean Exclusive OR;  
 M Complement of M;  
 → Transfer Into;  
 0 Bit = Zero;  
 00 Byte = Zero;

## CONDITION CODE SYMBOLS:

H Half-carry from bit 3;  
 I Interrupt mask  
 N Negative (sign bit)  
 Z Zero (byte)  
 V Overflow, 2's complement  
 C Carry from bit 7  
 R Reset Always  
 S Set Always  
 : Test and set if true, cleared otherwise  
 • Not Affected



TABLE 4 — INDEX REGISTER AND STACK MANIPULATION INSTRUCTIONS

|                      |          | BOOLEAN/ARITHMETIC OPERATION |   |   |        |   |   |       |   |   |       |   |   | COND. CODE REG. |   |   |                                                |   |   |   |   |   |   |
|----------------------|----------|------------------------------|---|---|--------|---|---|-------|---|---|-------|---|---|-----------------|---|---|------------------------------------------------|---|---|---|---|---|---|
| POINTER OPERATIONS   | MNEMONIC | IMMED                        |   |   | DIRECT |   |   | INDEX |   |   | EXTND |   |   | IMPLIED         |   |   | BOOLEAN/ARITHMETIC OPERATION                   |   |   |   |   |   |   |
|                      |          | OP                           | ~ | # | OP     | ~ | # | OP    | ~ | # | OP    | ~ | # | OP              | ~ | # | H                                              | I | N | Z | V | C |   |
| Compare Index Reg    | CPX      | 8C                           | 3 | 3 | 9C     | 4 | 2 | AC    | 6 | 2 | BC    | 5 | 3 |                 |   |   | $X_H - M, X_L - (M + 1)$                       | • | • | ⑦ | ↓ | ⑦ | • |
| Decrement Index Reg  | DEX      |                              |   |   |        |   |   |       |   |   |       |   |   | 09              | 4 | 1 | $X - 1 \rightarrow X$                          | • | • | • | ↓ | • | • |
| Decrement Stack Pntr | DES      |                              |   |   |        |   |   |       |   |   |       |   |   | 34              | 4 | 1 | $SP - 1 \rightarrow SP$                        | • | • | • | • | • | • |
| Increment Index Reg  | INX      |                              |   |   |        |   |   |       |   |   |       |   |   | 08              | 4 | 1 | $X + 1 \rightarrow X$                          | • | • | • | ↓ | • | • |
| Increment Stack Pntr | INS      |                              |   |   |        |   |   |       |   |   |       |   |   | 31              | 4 | 1 | $SP + 1 \rightarrow SP$                        | • | • | • | • | • | • |
| Load Index Reg       | LDX      | CE                           | 3 | 3 | DE     | 4 | 2 | EE    | 6 | 2 | FE    | 5 | 3 |                 |   |   | $M \rightarrow X_H, (M + 1) \rightarrow X_L$   | • | • | ⑨ | ↓ | R | • |
| Load Stack Pntr      | LDS      | 8E                           | 3 | 3 | 9E     | 4 | 2 | AE    | 6 | 2 | BE    | 5 | 3 |                 |   |   | $M \rightarrow SP_H, (M + 1) \rightarrow SP_L$ | • | • | ⑨ | ↓ | R | • |
| Store Index Reg      | STX      |                              |   |   | DF     | 5 | 2 | EF    | 7 | 2 | FF    | 6 | 3 |                 |   |   | $X_H \rightarrow M, X_L \rightarrow (M + 1)$   | • | • | ⑨ | ↓ | R | • |
| Store Stack Pntr     | STS      |                              |   |   | 9F     | 5 | 2 | AF    | 7 | 2 | BF    | 6 | 3 |                 |   |   | $SP_H \rightarrow M, SP_L \rightarrow (M + 1)$ | • | • | ⑨ | ↓ | R | • |
| Idx Reg → Stack Pntr | TXS      |                              |   |   |        |   |   |       |   |   |       |   |   | 35              | 4 | 1 | $X - 1 \rightarrow SP$                         | • | • | • | • | • | • |
| Stack Pntr → Idx Reg | TSX      |                              |   |   |        |   |   |       |   |   |       |   |   | 30              | 4 | 1 | $SP + 1 \rightarrow X$                         | • | • | • | • | • | • |

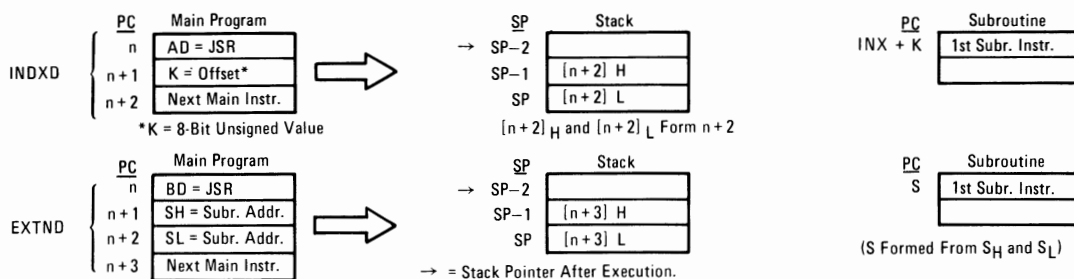
TABLE 5 — JUMP AND BRANCH INSTRUCTIONS

| OPERATIONS               | MNEMONIC | COND. CODE REG. |   |   |       |   |   |       |   |   |         |    |   | COND. CODE REG. |   |   |   |   |   |
|--------------------------|----------|-----------------|---|---|-------|---|---|-------|---|---|---------|----|---|-----------------|---|---|---|---|---|
|                          |          | RELATIVE        |   |   | INDEX |   |   | EXTND |   |   | IMPLIED |    |   | COND. CODE REG. |   |   |   |   |   |
|                          |          | OP              | ~ | # | OP    | ~ | # | OP    | ~ | # | OP      | ~  | # | COND. CODE REG. |   |   |   |   |   |
| Branch Always            | BRA      | 20              | 4 | 2 |       |   |   |       |   |   |         |    |   | 5               | 4 | 3 | 2 | 1 | 0 |
| Branch If Carry Clear    | BCC      | 24              | 4 | 2 |       |   |   |       |   |   |         |    |   | H               | I | N | Z | V | C |
| Branch If Carry Set      | BCS      | 25              | 4 | 2 |       |   |   |       |   |   |         |    |   | •               | • | • | • | • | • |
| Branch If = Zero         | BEQ      | 27              | 4 | 2 |       |   |   |       |   |   |         |    |   | •               | • | • | • | • | • |
| Branch If $\geq$ Zero    | BGE      | 2C              | 4 | 2 |       |   |   |       |   |   |         |    |   | •               | • | • | • | • | • |
| Branch If $>$ Zero       | BGT      | 2E              | 4 | 2 |       |   |   |       |   |   |         |    |   | •               | • | • | • | • | • |
| Branch If Higher         | BHI      | 22              | 4 | 2 |       |   |   |       |   |   |         |    |   | •               | • | • | • | • | • |
| Branch If $\leq$ Zero    | BLE      | 2F              | 4 | 2 |       |   |   |       |   |   |         |    |   | •               | • | • | • | • | • |
| Branch If Lower Or Same  | BLS      | 23              | 4 | 2 |       |   |   |       |   |   |         |    |   | •               | • | • | • | • | • |
| Branch If $<$ Zero       | BLT      | 2D              | 4 | 2 |       |   |   |       |   |   |         |    |   | •               | • | • | • | • | • |
| Branch If Minus          | BMI      | 2B              | 4 | 2 |       |   |   |       |   |   |         |    |   | •               | • | • | • | • | • |
| Branch If Not Equal Zero | BNE      | 26              | 4 | 2 |       |   |   |       |   |   |         |    |   | •               | • | • | • | • | • |
| Branch If Overflow Clear | BVC      | 28              | 4 | 2 |       |   |   |       |   |   |         |    |   | •               | • | • | • | • | • |
| Branch If Overflow Set   | BVS      | 29              | 4 | 2 |       |   |   |       |   |   |         |    |   | •               | • | • | • | • | • |
| Branch If Plus           | BPL      | 2A              | 4 | 2 |       |   |   |       |   |   |         |    |   | •               | • | • | • | • | • |
| Branch To Subroutine     | BSR      | 8D              | 8 | 2 |       |   |   |       |   |   |         |    |   | •               | • | • | • | • | • |
| Jump                     | JMP      |                 |   |   | 6E    | 4 | 2 | 7E    | 3 | 3 |         |    |   | •               | • | • | • | • | • |
| Jump To Subroutine       | JSR      |                 |   |   | AD    | 8 | 2 | BD    | 9 | 3 |         |    |   | •               | • | • | • | • | • |
| No Operation             | NOP      |                 |   |   |       |   |   |       |   |   | 02      | 2  | 1 | •               | • | • | • | • | • |
| Return From Interrupt    | RTI      |                 |   |   |       |   |   |       |   |   | 3B      | 10 | 1 | •               | • | • | • | • | • |
| Return From Subroutine   | RTS      |                 |   |   |       |   |   |       |   |   | 39      | 5  | 1 | •               | • | • | • | • | • |
| Software Interrupt       | SWI      |                 |   |   |       |   |   |       |   |   | 3F      | 12 | 1 | •               | • | • | • | • | • |
| Wait for Interrupt       | WAI      |                 |   |   |       |   |   |       |   |   | 3E      | 9  | 1 | •               | • | • | • | • | • |

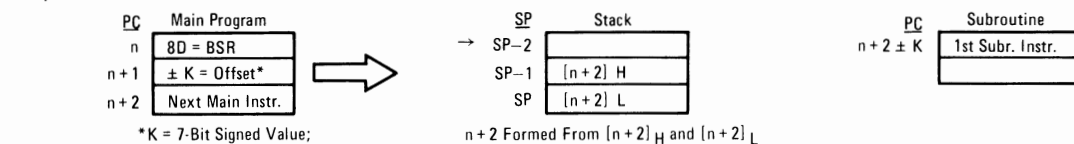


## SPECIAL OPERATIONS

## JSR, JUMP TO SUBROUTINE:



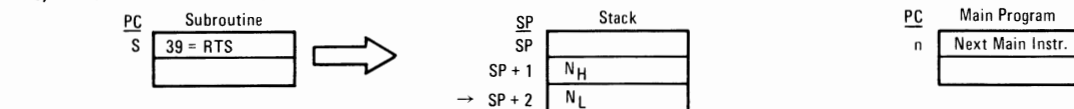
## BSR, BRANCH TO SUBROUTINE:



## JMP, JUMP:



## RTS, RETURN FROM SUBROUTINE:



## RTI, RETURN FROM INTERRUPT:

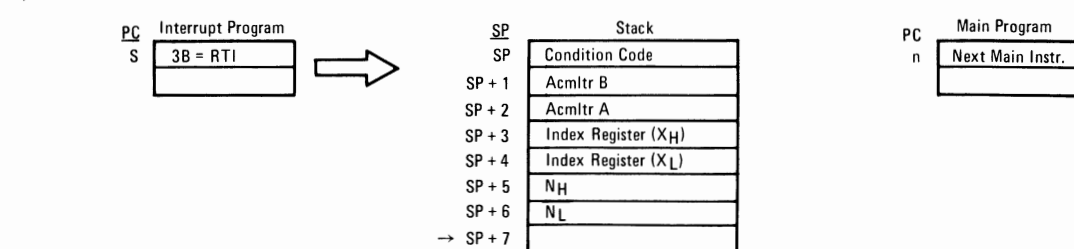


TABLE 6 – CONDITION CODE REGISTER MANIPULATION INSTRUCTIONS

| OPERATIONS                  | MNEMONIC | IMPLIED |   |   | BOOLEAN OPERATION   | COND. CODE REG. |   |   |   |   |   |
|-----------------------------|----------|---------|---|---|---------------------|-----------------|---|---|---|---|---|
|                             |          | OP      | ~ | # |                     | 5               | 4 | 3 | 2 | 1 | 0 |
|                             |          |         |   |   |                     | H               | I | N | Z | V | C |
| Clear Carry                 | CLC      | 0C      | 2 | 1 | $0 \rightarrow C$   | •               | • | • | • | • | R |
| Clear Interrupt Mask        | CLI      | 0E      | 2 | 1 | $0 \rightarrow I$   | •               | R | • | • | • | • |
| Clear Overflow              | CLV      | 0A      | 2 | 1 | $0 \rightarrow V$   | •               | • | • | • | R | • |
| Set Carry                   | SEC      | 0D      | 2 | 1 | $1 \rightarrow C$   | •               | • | • | • | • | S |
| Set Interrupt Mask          | SEI      | 0F      | 2 | 1 | $1 \rightarrow I$   | •               | S | • | • | • | • |
| Set Overflow                | SEV      | 0B      | 2 | 1 | $1 \rightarrow V$   | •               | • | • | • | S | • |
| Accmltr A $\rightarrow$ CCR | TAP      | 06      | 2 | 1 | $A \rightarrow CCR$ | 12              |   |   |   |   |   |
| CCR $\rightarrow$ Accmltr A | TPA      | 07      | 2 | 1 | $CCR \rightarrow A$ |                 |   |   |   |   |   |

## CONDITION CODE REGISTER NOTES:

(Bit set if test is true and cleared otherwise)

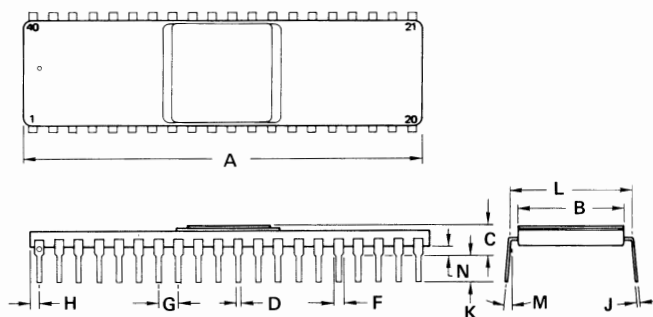
- 1 (Bit V) Test: Result = 10000000?
- 2 (Bit C) Test: Result = 00000000?
- 3 (Bit C) Test: Decimal value of most significant BCD Character greater than nine? (Not cleared if previously set.)
- 4 (Bit V) Test: Operand = 10000000 prior to execution?
- 5 (Bit V) Test: Operand = 01111111 prior to execution?
- 6 (Bit V) Test: Set equal to result of  $N \oplus C$  after shift has occurred.
- 7 (Bit N) Test: Sign bit of most significant (MS) byte = 1?
- 8 (Bit V) Test: 2's complement overflow from subtraction of MS bytes?
- 9 (Bit N) Test: Result less than zero? (Bit 15 = 1)
- 10 (All) Load Condition Code Register from Stack. (See Special Operations)
- 11 (Bit I) Set when interrupt occurs. If previously set, a Non-Maskable Interrupt is required to exit the wait state.
- 12 (All) Set according to the contents of Accumulator A.

## PIN ASSIGNMENT

|    |                 |                 |    |
|----|-----------------|-----------------|----|
| 1  | V <sub>SS</sub> | Reset           | 40 |
| 2  | Halt            | TSC             | 39 |
| 3  | $\phi 1$        | N.C.            | 38 |
| 4  | IRQ             | $\phi 2$        | 37 |
| 5  | VMA             | DBE             | 36 |
| 6  | NMI             | N.C.            | 35 |
| 7  | BA              | R/W             | 34 |
| 8  | V <sub>CC</sub> | D0              | 33 |
| 9  | A0              | D1              | 32 |
| 10 | A1              | D2              | 31 |
| 11 | A2              | D3              | 30 |
| 12 | A3              | D4              | 29 |
| 13 | A4              | D5              | 28 |
| 14 | A5              | D6              | 27 |
| 15 | A6              | D7              | 26 |
| 16 | A7              | A15             | 25 |
| 17 | A8              | A14             | 24 |
| 18 | A9              | A13             | 23 |
| 19 | A10             | A12             | 22 |
| 20 | A11             | V <sub>SS</sub> | 21 |

## PACKAGE DIMENSIONS

CASE 699-04



| DIM | MILLIMETERS |       | INCHES    |       |
|-----|-------------|-------|-----------|-------|
|     | MIN         | MAX   | MIN       | MAX   |
| A   | 50.04       | 51.05 | 1.970     | 2.010 |
| B   | 13.46       | 14.22 | 0.530     | 0.560 |
| C   | 3.05        | 3.94  | 0.120     | 0.155 |
| D   | 0.38        | 0.51  | 0.015     | 0.020 |
| F   | 0.89        | 1.40  | 0.035     | 0.055 |
| G   | 2.54 BSC    |       | 0.100 BSC |       |
| H   | 0.89        | 1.40  | 0.035     | 0.055 |
| J   | 0.20        | 0.28  | 0.008     | 0.011 |
| K   | 3.05        | 3.68  | 0.120     | 0.145 |
| L   | 14.86       | 15.87 | 0.585     | 0.625 |
| M   | 15°         |       | 15°       |       |
| N   | 0.51        | 1.14  | 0.020     | 0.045 |

## NOTES:

1. LEADS WITHIN 0.13 mm (0.005) RADIUS OF TRUE POSITION AT SEATING PLANE, AT MAXIMUM MATERIAL CONDITION.
2. DIMENSION "L" TO INSIDE OF LEADS (MEASURED 0.51 mm (0.020) BELOW PACKAGE BASE)



TABLE 7 — INSTRUCTION ADDRESSING MODES AND ASSOCIATED EXECUTION TIMES  
(Times in Machine Cycles)

|     | (Dual Operand) | ACCX | Immediate | Direct | Extended | Indexed | Implied | Relative |     | (Dual Operand) | ACCX | Immediate | Direct | Extended | Indexed | Implied |
|-----|----------------|------|-----------|--------|----------|---------|---------|----------|-----|----------------|------|-----------|--------|----------|---------|---------|
| ABA |                | •    | •         | •      | •        | •       | 2       | •        | INC | 2              | •    | •         | •      | 6        | 7       | •       |
| ADC | x              | •    | 2         | 3      | 4        | 5       | •       | •        | INS | •              | •    | •         | •      | •        | •       | 4       |
| ADD | x              | •    | 2         | 3      | 4        | 5       | •       | •        | INX | •              | •    | •         | •      | •        | •       | 4       |
| AND | x              | •    | 2         | 3      | 4        | 5       | •       | •        | JMP | •              | •    | •         | 3      | 4        | •       | •       |
| ASL |                | 2    | •         | •      | 6        | 7       | •       | •        | JSR | •              | •    | •         | 9      | 8        | •       | •       |
| ASR |                | 2    | •         | •      | 6        | 7       | •       | •        | LDA | x              | 2    | 3         | 4      | 5        | •       | •       |
| BCC |                | •    | •         | •      | •        | •       | •       | 4        | LDS | •              | 3    | 4         | 5      | 6        | •       | •       |
| BCS |                | •    | •         | •      | •        | •       | •       | 4        | LDX | •              | 3    | 4         | 5      | 6        | •       | •       |
| BEA |                | •    | •         | •      | •        | •       | •       | 4        | LSR | 2              | •    | •         | 6      | 7        | •       | •       |
| BGE |                | •    | •         | •      | •        | •       | •       | 4        | NEG | 2              | •    | •         | 6      | 7        | •       | •       |
| BGT |                | •    | •         | •      | •        | •       | •       | 4        | NOP | •              | •    | •         | •      | •        | •       | 2       |
| BHI |                | •    | •         | •      | •        | •       | •       | 4        | ORA | x              | 2    | 3         | 4      | 5        | •       | •       |
| BIT | x              | •    | 2         | 3      | 4        | 5       | •       | •        | PSH | •              | •    | •         | •      | •        | •       | 4       |
| BLE |                | •    | •         | •      | •        | •       | •       | 4        | PUL | •              | •    | •         | •      | •        | •       | 4       |
| BLS |                | •    | •         | •      | •        | •       | •       | 4        | ROL | 2              | •    | •         | 6      | 7        | •       | •       |
| BLT |                | •    | •         | •      | •        | •       | •       | 4        | ROR | 2              | •    | •         | 6      | 7        | •       | •       |
| BMI |                | •    | •         | •      | •        | •       | •       | 4        | RTI | •              | •    | •         | •      | •        | •       | 10      |
| BNE |                | •    | •         | •      | •        | •       | •       | 4        | RTS | •              | •    | •         | •      | •        | •       | 5       |
| BPL |                | •    | •         | •      | •        | •       | •       | 4        | SBA | •              | •    | •         | •      | •        | •       | 2       |
| BRA |                | •    | •         | •      | •        | •       | •       | 4        | SBC | x              | 2    | 3         | 4      | 5        | •       | •       |
| BSR |                | •    | •         | •      | •        | •       | •       | 8        | SEC | •              | •    | •         | •      | •        | •       | 2       |
| BVC |                | •    | •         | •      | •        | •       | •       | 4        | SEI | •              | •    | •         | •      | •        | •       | 2       |
| BVS |                | •    | •         | •      | •        | •       | •       | 4        | SEV | •              | •    | •         | •      | •        | •       | 2       |
| CBA |                | •    | •         | •      | •        | •       | 2       | •        | STA | x              | •    | •         | 4      | 5        | 6       | •       |
| CLC |                | •    | •         | •      | •        | •       | 2       | •        | STS | •              | •    | •         | 5      | 6        | 7       | •       |
| CLI |                | •    | •         | •      | •        | •       | 2       | •        | STX | •              | •    | •         | 5      | 6        | 7       | •       |
| CLR |                | 2    | •         | •      | 6        | 7       | •       | •        | SUB | x              | •    | 2         | 3      | 4        | 5       | •       |
| CLV |                | •    | •         | •      | •        | •       | 2       | •        | SWI | •              | •    | •         | •      | •        | •       | 12      |
| CMP | x              | •    | 2         | 3      | 4        | 5       | •       | •        | TAB | •              | •    | •         | •      | •        | •       | 2       |
| COM |                | 2    | •         | •      | 6        | 7       | •       | •        | TAP | •              | •    | •         | •      | •        | •       | 2       |
| CPX |                | •    | 3         | 4      | 5        | 6       | •       | •        | TBA | •              | •    | •         | •      | •        | •       | 2       |
| DAA |                | •    | •         | •      | •        | •       | 2       | •        | TPA | •              | •    | •         | •      | •        | •       | 2       |
| DEC |                | 2    | •         | •      | 6        | 7       | •       | •        | TST | 2              | •    | •         | 6      | 7        | •       | •       |
| DES |                | •    | •         | •      | •        | •       | 4       | •        | TSX | •              | •    | •         | •      | •        | •       | 4       |
| DEX |                | •    | •         | •      | •        | •       | 4       | •        | TSX | •              | •    | •         | •      | •        | •       | 4       |
| EOR | x              | •    | 2         | 3      | 4        | 5       | •       | •        | WAI | •              | •    | •         | •      | •        | •       | 9       |

NOTE: Interrupt time is 12 cycles from the end of the instruction being executed, except following a WAI instruction. Then it is 4 cycles.

